

a textbook of current ideas in physics, *Institutions de physique*, which included her own work, and translated the whole of Isaac Newton's *Principia Mathematica* into French. She died in childbirth, with Voltaire and a younger lover by her side.

GEORGES-LOUIS LECLERC

1707–1788

The Comte de Buffon, Georges-Louis Leclerc, caused a stir when he noted the similarities between humans and apes and suggested that living things could be modified by environmental changes such as migration, leading to similar species existing in different places. He also argued that Earth was much older than biblical explanations allowed. Inspired by his work at the Jardin du Roi (the royal botanical gardens in Paris), the French naturalist tried to explain all of nature in his encyclopedia of natural history, but he only published 36 of the proposed 50 volumes before his death.

JOSEPH BLACK

1728–1799

British chemist and physician Joseph Black was born in Bordeaux, France, and studied the arts, then medicine, at the University of Glasgow, Scotland. As part of his 1754 doctoral thesis, Black discovered carbon dioxide, which he called “fixed air.” After continuing his studies in Edinburgh, he returned to Glasgow to teach chemistry and began his experiments on the effects of heat on liquids. He found that once a pan of water on a stove reaches boiling point, the temperature of the water stops rising and the heat instead turns the water into vapor. He called this “latent” (hidden) heat. Black rarely published his findings, preferring to announce them to his students during his lectures.

JAN INGENHOUSZ

1730–1799

In 1779, Dutch-born physician Jan Ingenhousz submerged a plant in water and watched as oxygen bubbles formed on the underside of its leaves. He discovered that oxygen formed only on the green parts of the plant that were exposed to sunlight (in a process now known as photosynthesis). He also found that in the dark, plants emit low levels of carbon dioxide. Ingenhousz worked as a doctor in London, England, and was an early advocate of vaccination against smallpox. He even traveled to Vienna to inoculate the Austrian empress Maria Theresa and her family.

JEAN-BAPTISTE LAMARCK

1744–1829

After an injury forced him to leave the French military, Jean-Baptiste Lamarck went to work at the royal botanical gardens in Paris. There, he wrote *Flore Française*, a book about plants. When the gardens became a natural history museum, Lamarck studied invertebrates and came up with the first major theory of evolution. He suggested that over time, living things tend to progress, and that organisms change in response to their environment (for example, by growing longer legs to wade in water). These acquired characteristics are passed on to their offspring. Lamarck's theory was unpopular, and he died in poverty.

PIERRE-SIMON LAPLACE

1749–1827

Born in Normandy, France, Pierre-Simon Laplace developed an interest in mathematics while studying theology in Caen. He moved to Paris, where he secured a teaching position at the Ecole

Militaire. In 1773, he began his major work: proving that gravity will keep the solar system stable in the long run and that the self-correcting “perturbations” noted by Isaac Newton (and put down to divine intervention) are, in fact, also made by gravity. As well as proving Newton's theory, Laplace came up with the idea that the solar system formed from a rotating cloud of hot gases.

CHRISTIAN KONRAD SPRENGEL

1750–1816

German botanist Christian Konrad Sprengel achieved little recognition in his lifetime. It was not until 1841 that his work on how fertilization occurs in plants was brought to light by Charles Darwin. Sprengel worked as a teacher in Spandau, Prussia (now Germany), but was dismissed for spending too much time studying plants. He moved to Berlin and, in 1793, published his observations, including the fact that insects carry pollen from the male parts of flowers to the female parts, and that these insects are attracted by the color of a flower's nectaries (nectar-producing organs).

JOSEPH FOURIER

1768–1830

Before settling on a career as a mathematician, Joseph Fourier trained as a priest, dabbled in politics, and traveled with Napoleon to Egypt to study Egyptian relics. The Frenchman made his name with his study of heat flow, in which he showed that a wave of any shape can be represented by adding together simpler waves called sines and cosines. Fourier analysis is now used in many areas, including electronics. He also identified the greenhouse effect, realizing that Earth is warm because gases in its atmosphere trap heat from the Sun and prevent it from escaping.